

Fibonacci In Human Life

Life is based on patterns, repetitions, but also randoms at the same time. This sparks an idea of trying to find a “secret code” for the pattern of the life, which makes life both rely on patterns but stay completely random at the same time.

Fibonacci sequence is found by Leonardo Bonacci, who wrote the Liber Abaci in 1202. The Fibonacci sequence is an ordered sequence of numbers that begins with 1 and progresses by adding the previous number to the current one, continuing eternally. The larger the number gets, the division of the number with previous gets closer to 1.618, which is also called *Phi*, *The Golden Ratio*.

The Golden Ratio can be observed in nature, particularly in the arrangement of flower petals. Flowers need to align their petals to maximize light absorption, and they achieve this by growing them based on the golden angle, calculated using specific equations.

For instance, the equation $(1 - (1/1.618)) \approx 1 - 0.618 = 0.382$, and multiplying 0.382 by 360 gives approximately 137.5 degrees. This angle optimizes the number of petals while minimizing the blocked area caused by overlapping petals. Although flowers don't consciously solve this mathematical problem, nature has ingeniously determined that growing petals at this angle maximizes light absorption and enhances photosynthesis efficiency.

Fibonacci is also frequently used in reading graphs. For instance, in stocks, cryptocurrency, and forex trading, most investors rely on graphs to make informed decisions about whether to invest their money in a particular asset. The Fibonacci Retracement tool is one of the most commonly used tools in technical analysis.

The Fibonacci Retracement tool comprises a series of numbers beginning at 0 and extending up to 1. These numbers are 0, 0.236, 0.382, 0.5, 0.618, 0.786, and 1. Each number represents a significant and unique signal about the price movement in trading.

- The signal of 0.786 suggests that the price movement might stabilize or reverse at that point.
- 0.618 is generally considered in conjunction with 0.5. The area between 0.5 and 0.618 is referred to as The Golden Area. This is because it signals that the trend might continue after passing through that area, while both 0.5 and 0.618 indicate that movement might stabilize at that point.
- The signal of 0.382 indicates a high probability that the price movement will stabilize or reverse at that point.
- The signal of 0.236 indicates a slight possibility that the price movement might stabilize or reverse at that point.
- The numbers 0 and 1 signify the commencement and conclusion of the price movement.

While the Fibonacci sequence can be applied to analyzing asset graphs, this essay delves into explaining why it can be applied to analyzing asset graphs, and its application in real-world graphs such as global birth rates, deaths caused by COVID-19 statistics, and individual graphs like screen time usage. Through this study, I aim to demonstrate that the Fibonacci sequence actually manifests even in seemingly random and unplanned aspects of our lives.

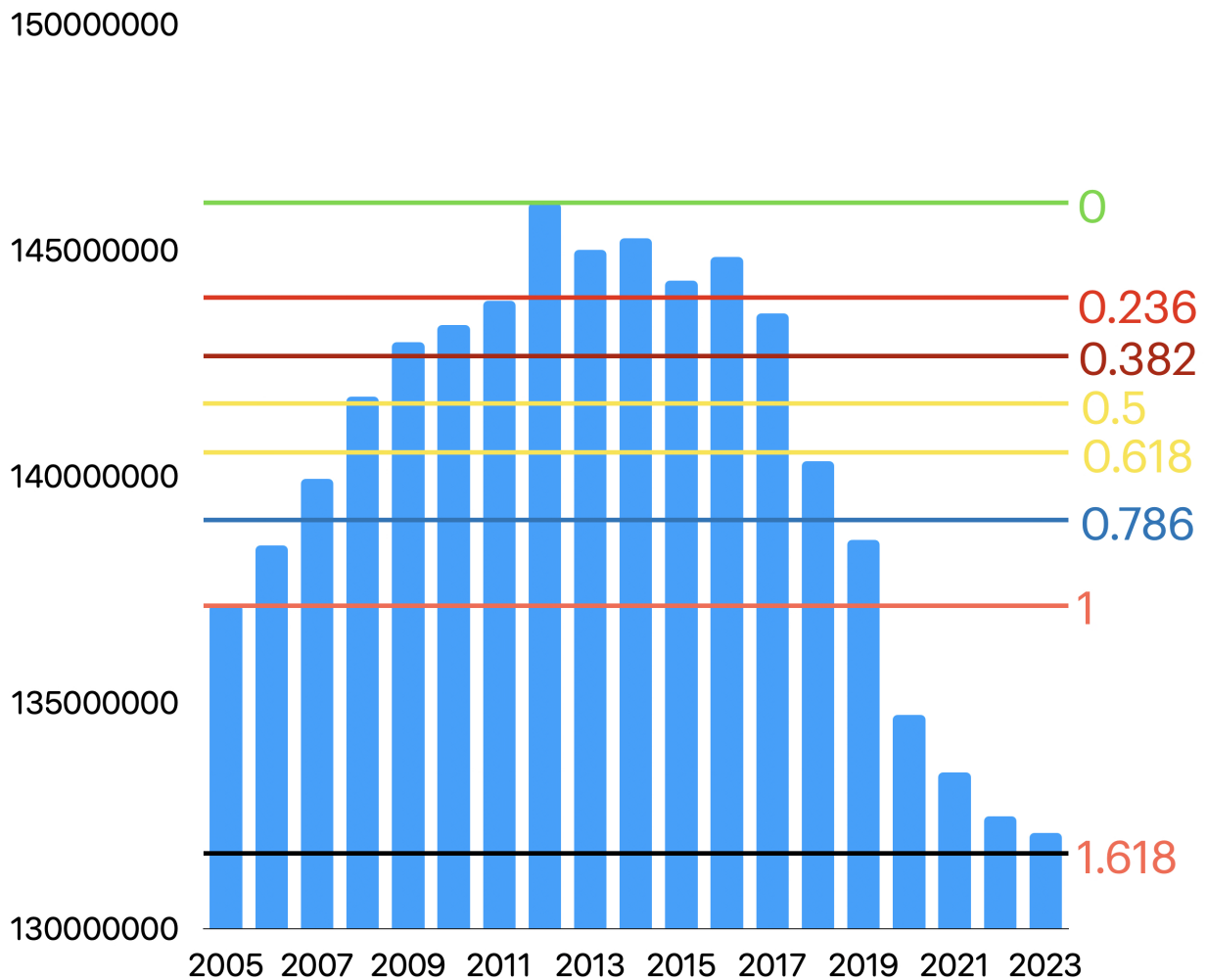
Before continuing, I want to explain what truly defines an asset's price. It is not its intrinsic mathematical value, but the demand people place upon it. When enough people believe an asset is worth a certain price, it becomes worth that price and when confidence fades, so does the value of the asset. This means that every movement on a price chart, every bounce at a Fibonacci level, is a reflection of a collective human decision. When a company faces bad news, it isn't the bad news that decreases the price of the assets, but the trust and demand of people to that company. Fear, trust, patience, and greed are what move markets. This realization is the foundation of the argument presented in this essay: if Fibonacci retracement accurately maps the behavior of asset prices, and asset prices are driven entirely by human emotion and sentiment, then it is not just a financial tool. It is a map of human psychology. And if that is true, it follows that the same patterns should be observable anywhere human emotion and sentiment govern behavior. Daily life, governed by the same impulses of motivation, fatigue, patience, and habit, may be no exception.

While it is tempting to dismiss Fibonacci retracement levels as merely horizontal lines dressed in mathematical clothing, this argument misses a more fundamental question: why these specific ratios, and not others? Institutions did not gather in a room and arbitrarily agree to watch 61.8%. These levels were adopted precisely because they already appeared to reflect something real about how human behavior moves, how far a crowd pushes before exhaustion sets in, how deep confidence must fall before it rebuilds, how long momentum sustains before psychology reverses it. The widespread adoption of Fibonacci in trading is not the cause of the pattern, but a consequence of it. Traders recognized something in these ratios before algorithms enforced them. This distinction matters enormously for the argument presented in this essay. If Fibonacci levels only worked because millions of traders simultaneously watched them, they would be useless outside of financial markets, and yet, as the birth rate and COVID data demonstrate, the patterns persist in domains where no trader is present and no algorithm is running. This suggests that what Fibonacci retracement actually measures is not market mechanics, but the underlying rhythm of human behavior itself. The market did not create this pattern. It simply revealed it first.

Academic literature consistently confirms that markets run on human psychology, not pure logic. Baker and Wurgler state that "the question is no longer whether investor sentiment affects stock prices, but how to measure investor sentiment and quantify its effects."¹ Robert Shiller, a Nobel Prize-winning economist, argues that "irrational exuberance is the psychological basis of a speculative bubble."²

¹ Baker, M., & Wurgler, J. (2007). Investor Sentiment in the Stock Market. *Journal of Economic Perspectives*, 21(2), 129–152.

² Shiller, R. J. (2000). *Irrational Exuberance*. Princeton University Press.



Global Birth Rate From 2005 To 2023 (Image 1.0.1)

Here's a bar graph depicting the global birth rate from 2005 to 2023, with Fibonacci retracement lines drawn on top (Image 1.0.1).

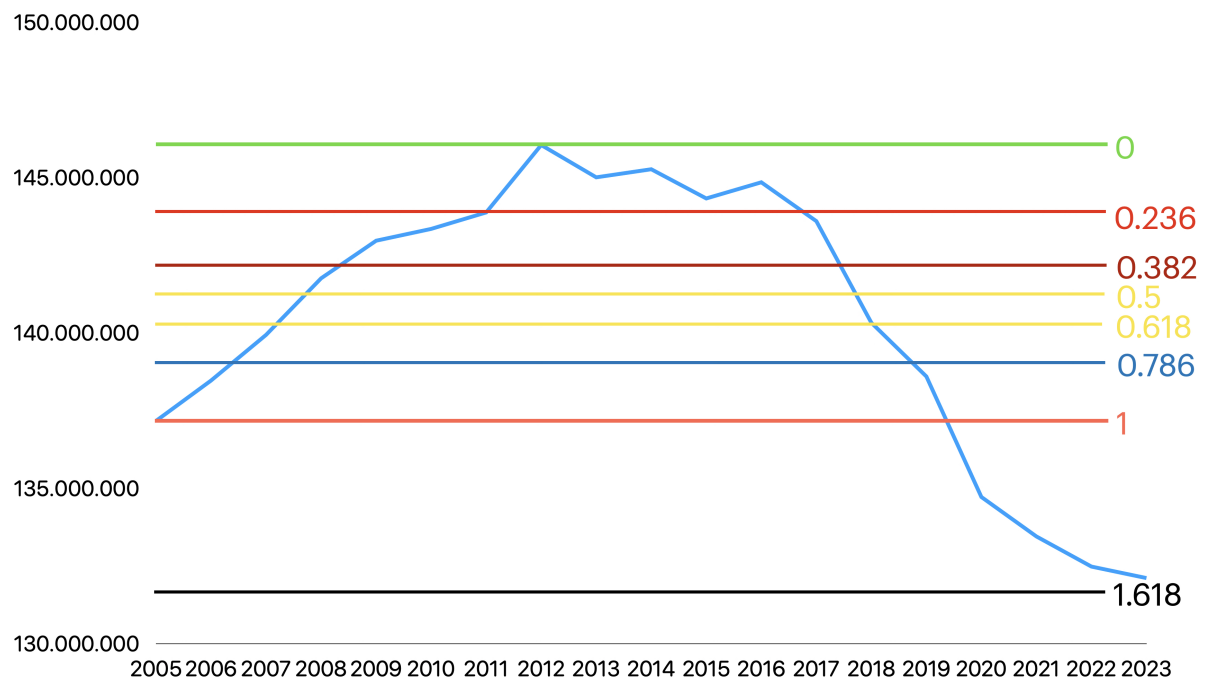
The global birth rate is affected by various problems such as global wars, viruses, conditions, economical situations, so one would expect it to have no mathematical pattern at all.

It could be observed that the Fibonacci retracement line is drawn from the beginning of the graph, in 2005, to the peak of the largest bar.

The global birth rate has been steadily increasing from 2005 to 2013. However, after reaching its peak in 2013, it gradually began to decline. In 2017, a sharp

downward trend commenced, which persisted until 2021. The trend started to decelerate between 2021 and 2023.

Viewing the same data as a line graph would be more helpful in commenting on the Fibonacci retracement.



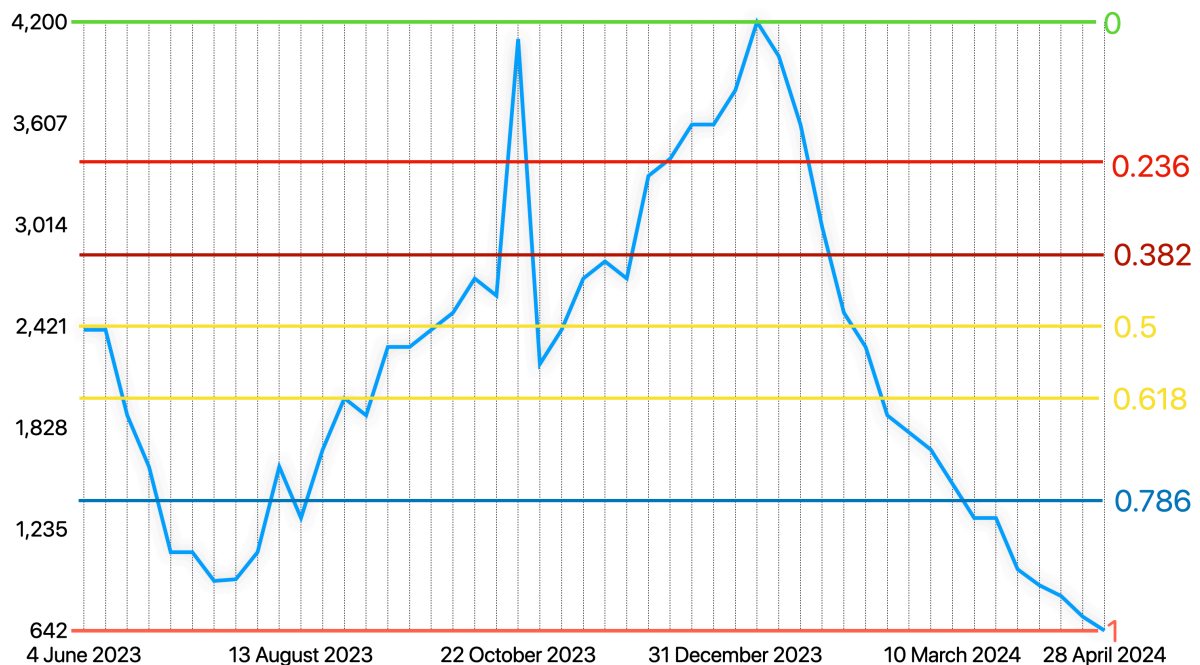
Global Birth Rate From 2005 To 2023 (Image 1.0.2)

Here's the line graph of the global birth rate from 2005 to 2023, with Fibonacci retracement lines drawn on top (Image 1.0.2).

Let's comment on the graph as a trader would comment on an asset price. The trend line has crossed the 0.5 and 0.618 levels, also known as the golden area. This indicates a strong upward trend between 2007 and 2008, consistent with the signal of the golden area, suggesting further growth. On the other hand, the trend line between 2017 and 2018 crossed the golden area with a strong downward trend and continued moving downward, also consistent with the signal of the golden area.

It's also evident that in 2011, when the data reached the 0.236 Fibonacci level, the upward movement gained momentum. Conversely, in 2019, at the 0.786 Fibonacci level, the downward movement gained strength.

The global birth rate graph challenges the idea of global/human life situations not having a pattern, and happening randomly. Furthermore, Fibonacci patterns could be seen in not only global birth rate, but various global data too.

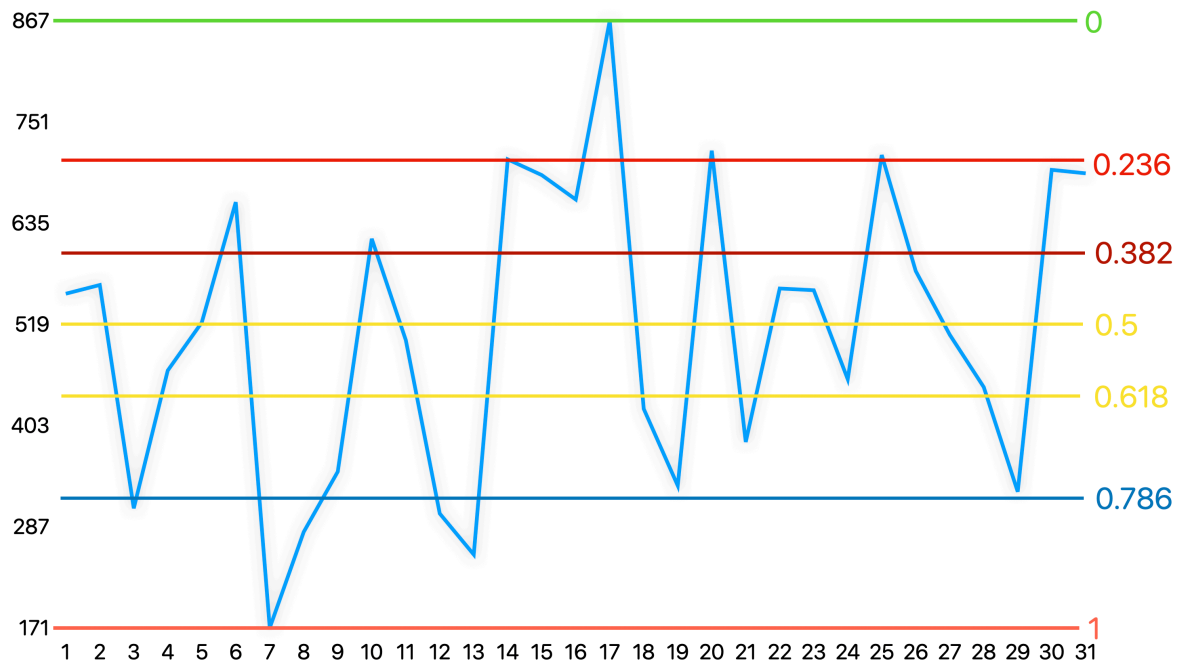


Weekly Deaths Caused By COVID-19 From 4 June 2023 To 28 April 2024 (Image 1.1)

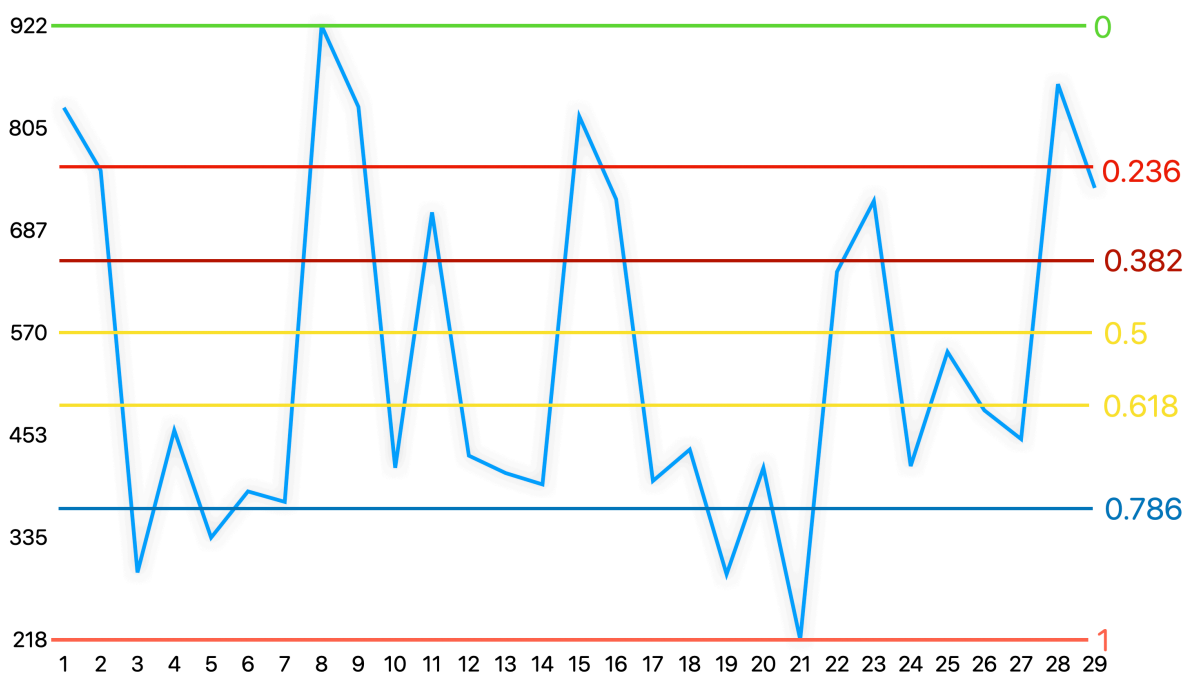
This graph shows weekly deaths caused by COVID-19 from 4 June 2023 To 28 April 2024 (Image 1.1). If we were to talk about the movements of the chart, it could be said that it shows tendency to align with the Fibonacci retracement; bounces from the confident levels, tests and breaks levels and follows a simple trend movement of rise and fall all through the graph.

These graphs [Global Birth Rate (Image 1.0.1, Image 1.0.2), Weekly Deaths Caused By COVID-19 (Image 1.1)] signal that Fibonacci might have a significant effect on the human life and global patterns as in the nature.

Besides global and complete patterns showing tendency to align with Fibonacci, the idea of effect of Fibonacci in individual life is the most essential point of Fibonacci in human life.



Daily Screen Time In Minutes Throughout May 2025 (Image 1.2.1)



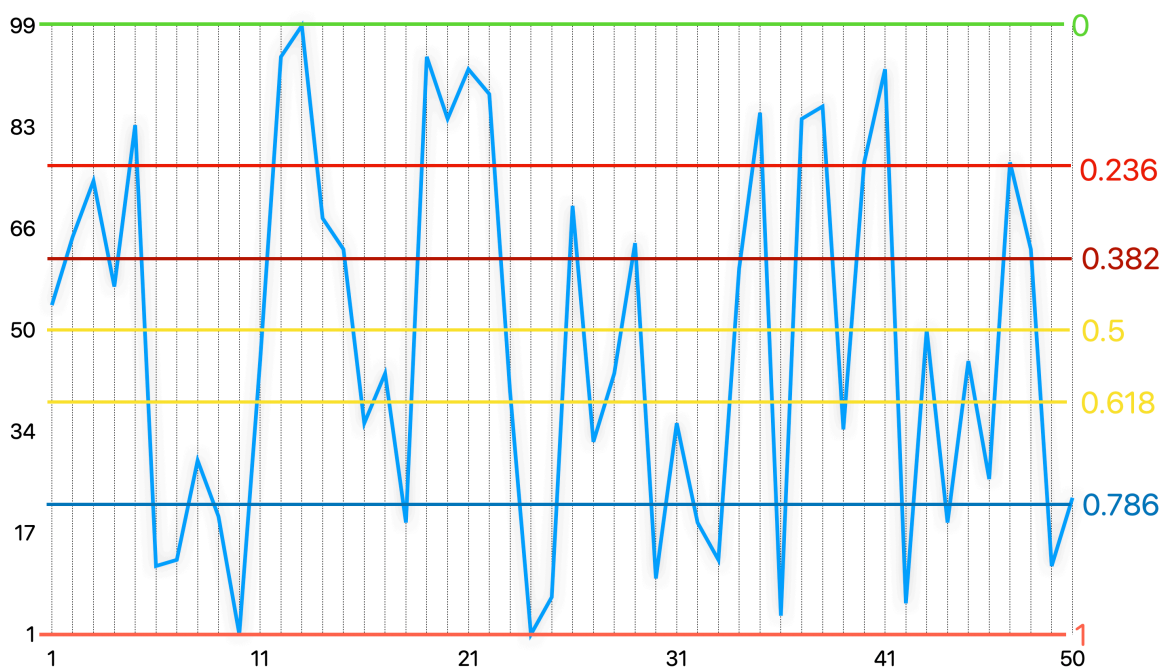
Daily Screen Time In Minutes From 14 February 2026 To 14 March 2026 (Image 1.2.2)

These are graphs of the amount of time I spent on my computer each day throughout May 2025 (Image 1.2.1) and from February 14th to March 14th (Image 1.2.2) converted into minutes and put into a Fibonacci retracement. I ran this experiment and gathered data at the end of the month without forcing myself to neither use nor shut down my computer, and to make the experiment more fair and objective, I didn't check the data until the end of the experiment.

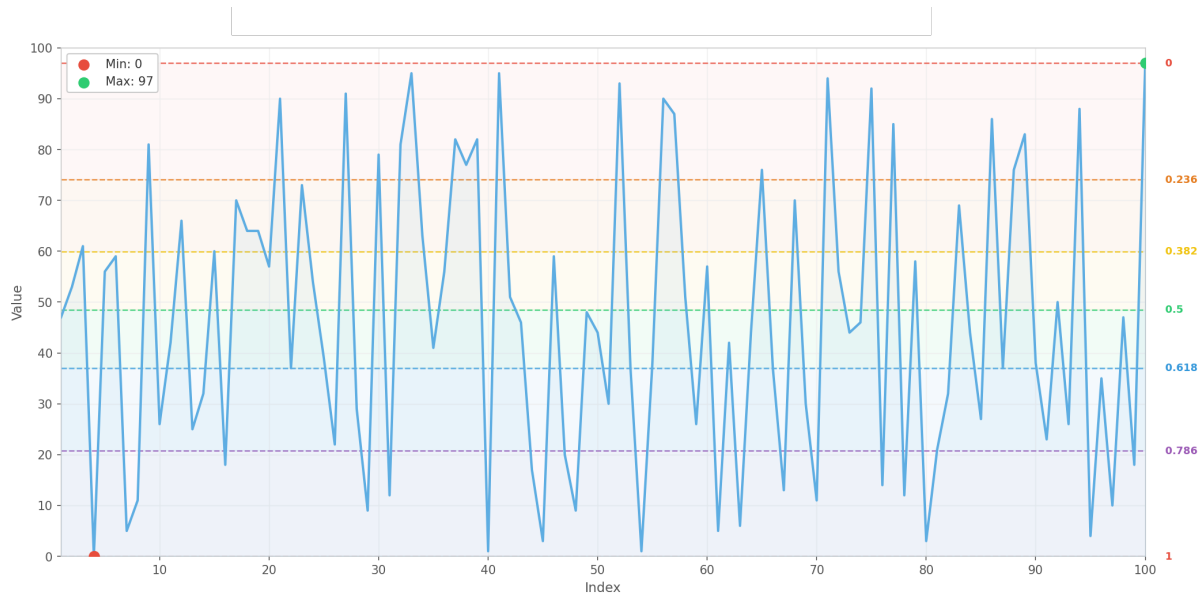
These durations are fully random as they include days like when I wasn't at home, when I had to go outside and shut my computer down immediately, when I was at home all day, and when I was at school.

To review the graphs, it could be said that the 0.786 and 0.236 Fibonacci levels are the key levels of my own brain's anchor as the duration of work I do on my computer keeps falling down to the 0.786 level but rarely crosses that level, often bouncing back up meanwhile it keeps increasing up to the 0.236 level but rarely crosses that level.

It could be argued drawing any horizontal line on any graph would seem like the graph movement is using that line as a support/bounce back level, ignoring the data category, type or amount.



50 Random Numbers In A Scale Of 0-100 From A Python Script (Image 1.3.1) (see Appendix A)



100 Random Numbers In A Scale Of 0-100 From A Python Script (Image 1.3.2) (see Appendix B)

Here are two graphs of random numbers in a scale of 0-100 created by a Python script and Fibonacci retracements drawn on them (Image 1.3.1, Image 1.3.2).

These graphs support the idea that Fibonacci doesn't appear in every random data, but only in those that are related with nature, as the Fibonacci levels aren't as quite working as the support/bounce back levels as they do in nature and daily life based graphs, even when the data quantity is increased for slightly more chance [50 Random Numbers (Image 1.3.1), 100 Random Numbers (Image 1.3.2)], causing the graph to appear fully random, no signs of any pattern.

The data presented in this essay, ranging from global birth rates to weekly COVID deaths and individual screen time, consistently shows tendency to align with Fibonacci in domains where no trader, no algorithm, and no financial incentive exists. This suggests that what Fibonacci retracement actually maps is not price or value, but the behavioral rhythm underneath it. Motivation builds, exhaustion sets in, confidence recovers, and fear takes over and these cycles, whether they play out in a market, a population, or a single person's daily habits, appear to follow proportions that nature has been using long before humans discovered them. Whether this is a deep mathematical truth about how energy and behavior move through systems, or simply a reflection of the consistent psychological limits we all share, remains an open question. But the Fibonacci sequence may be less of a financial tool and more of a mirror that reflects the hidden structure of human life back at us, even when we are not looking for it.

References

Baker, M., & Wurgler, J. (2007). Investor Sentiment in the Stock Market. *Journal of Economic Perspectives*, 21(2), 129–152.

Shiller, R. J. (2000). *Irrational Exuberance*. Princeton University Press.

Image 1.0.1, Image 1.0.2 - Global Birth Rate: <https://ourworldindata.org/grapher/births-and-deaths-projected-to-2100?time=earliest..2100>

Image 1.1 - Weekly Deaths Caused by COVID-19: <https://data.who.int/dashboards/covid19/deaths>

Appendices

Appendix A — Python Script: Random Number Graph (50 values, in a scale of 0-100):

```
import random

# Generates 50 random integers between 0 and 100
numbers = [random.randint(0, 100) for _ in range(50)]
```

Appendix B — Python Script: Random Number Graph (100 values, in a scale of 0-100):

```
import random

# Generates 100 random integers between 0 and 100
numbers = [random.randint(0, 100) for _ in range(100)]
```